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COIL COMPONENT

Abstract

In a coil component, each of a plurality of coil conductors is connected at an end thereof to another coil conductor of the plurality of coil conductors. The plurality of coil conductors include a first coil conductor and a second coil conductor. The first coil conductor extends along a Y-axis direction. The second coil conductor is connected to the first coil conductor. The second coil conductor extends from the first coil conductor toward a second end surface such that the shortest distance from a side surface to the second coil conductor increases as the second coil conductor approaches the second end surface. When viewed along a second direction orthogonal to a direction of a coil axis of a coil and a first direction, the first coil conductor does not overlap a portion which is closest to the first coil conductor in the second side surface electrode portion.

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Background/Summary

TECHNICAL FIELD The present invention relates to a coil component.
BACKGROUND
[0001] A known coil component includes an element body, an external electrode provided on the element body, and a coil disposed inside the element body (for example, JP2018-56513). The coil is connected to the external electrode. The coil includes a plurality of coil conductors.
SUMMARY
[0002] In the above-described coil component, in order to secure inductance, it is conceivable to increase the area of a region surrounded by the coil when viewed along a coil axis direction. In the coil components having the same size, as the area of the region is larger, the shortest distance between the coil and the external electrode is smaller. The inventor of the present application has focused on a phenomenon in which a Q value decreases when an electric signal in a relatively high frequency band conducts through a coil even if a desired Q value is secured when an electric signal in a relatively low frequency band conducts through the coil.
[0003] An object of one aspect of the present invention is to provide a coil component capable of securing both inductance and a Q value for an electric signal in a relatively high frequency band while achieving compactness.
[0004] As a result of further intensive studies, the inventor of the present application has found that when there is a position where a potential difference between the coil and the external electrode is relatively large and the distance is relatively short, the Q value decreases for an electric signal having a relatively low frequency. A coil component according to one aspect of the present invention includes an element body, a first external electrode, a second external electrode, and a coil. The element body includes first and second end surfaces opposite to each other in a first direction, and a side surface connecting the first and second end surfaces. The first external electrode includes a first end surface electrode portion provided on the first end surface, and a first side surface electrode portion provided on the side surface and extending from the first end surface toward the second end surface. The second external electrode includes a second end surface electrode portion provided on the second end surface, and a second side surface electrode portion provided on the side surface and extending from the second end surface toward the first end surface. The coil is disposed inside the element body. The coil is connected to the first end surface electrode portion at the first end surface, and is connected to the second end surface electrode portion at the second end surface. The coil includes a plurality of coil conductors. Each of the plurality of coil conductors includes a pair of ends, and is connected at the end to another coil conductor of the plurality of coil conductors. The plurality of coil conductors include a first coil conductor and a second coil conductor. The first coil conductor extends along the first direction. The second coil conductor is connected to the first coil conductor. The second coil conductor extends from the first coil conductor toward the second end surface such that the shortest distance from the side surface to the second coil conductor increases as the second coil conductor approaches the second end surface. When viewed along a second direction orthogonal to a coil axis direction of the coil and the first direction, the first coil conductor does not overlap a portion which is closest to the first coil conductor in the second side surface electrode portion.
[0005] In this coil component, since the first coil conductor connected to the first external electrode is separated from the second external electrode as viewed along the second direction, the Q value

can be secured even for a signal in a relatively high frequency band while achieving compactness. Since the second coil conductor is connected to the first coil conductor, and extends from the first coil conductor toward the second end surface such that the shortest distance from the side surface to the second coil conductor increases as the second coil conductor approaches the second end surface, the area of a region surrounded by the coil can be secured when viewed along the coil axis direction. As a result, it is possible to secure inductance while achieving compactness.
[0006] In the one aspect, the coil may further include a third coil conductor connected to the end different from the end connected to the first coil conductor, in the pair of ends of the second coil conductor. The third coil conductor may extend along the second direction. In this case, when viewed along the coil axis direction, a larger area of the region surrounded by the coil can be secured. As a result, it is possible to more reliably secure inductance while achieving compactness.
[0007] In the one aspect, a portion of the coil may overlap the second side surface electrode portion when viewed along the second direction. In this case, when viewed along the coil axis direction, a larger area of the region surrounded by the coil can be secured. As a result, it is possible to more reliably secure inductance while achieving compactness.
[0008] In the one aspect, the region surrounded by the plurality of coil conductors may have an octagonal shape when viewed along the coil axis direction. In this case, when viewed along the coil axis direction, a larger area of the region surrounded by the coil can be secured. As a result, it is possible to more reliably secure inductance while achieving compactness.
Advantageous Effects of Invention
[0009] One aspect of the present invention provides a coil component capable of securing both inductance and a Q value for an electric signal in a relatively high frequency band while achieving compactness.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0010] FIG. **1** is a perspective view of a coil component in an embodiment;
[0011] FIG. **2** is a partial cross-sectional view of the coil component in a YZ-axis plane;
[0012] FIG. **3** is a partial cross-sectional view of the coil component in an XY-axis plane;
[0013] FIG. **4** is a partial cross-sectional view of the coil component in an XZ-axis plane;
[0014] FIG. **5** is a partial cross-sectional view of a coil component in the YZ-axis plane in a modification of the embodiment;
[0015] FIG. **6** is a partial cross-sectional view of the coil component in the XY-axis plane;
[0016] FIG. **7** is a partial cross-sectional view of the coil component in the XZ-axis plane;
[0017] FIGS. **8**A to **8**D illustrate a coil conductor in each layer; and
[0018] FIG. **9** is a graph illustrating frequency characteristics of a Q value.
DETAILED DESCRIPTION
[0019] Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings. In the description of the drawings, the same reference numerals are used for the same or equivalent elements, and redundant description will be omitted.
[0020] First, a coil component in the present embodiment will be described with reference to FIGS. **1** to **4**. The coil component **1** is, for example, a multilayer coil component. FIG. **1** is a perspective view of the coil component in the present embodiment. FIG. **2** is a partial cross-sectional view of the coil component in a YZ-axis plane. FIG. **3** is a partial cross-sectional view of the coil component in an XY-axis plane. FIG. **4** is a partial cross-sectional view of the coil component in an XZ-axis plane.
[0021] As illustrated in FIG. **1**, the coil component **1** includes an element body **2** and external electrodes **5** and **6**. For example, when the external electrode **5** is a first external electrode, the

external electrode **6** corresponds to a second external electrode. The coil component **1** is solder-mounted to an electronic device, for example. The electronic device includes, for example, a circuit board or an electronic component. In the present embodiment, the element body **2** is formed by a plurality of element body layers stacked in an X-axis direction.

[0022] The element body **2** has, for example, an insulating property. The element body **2** is made of, for example, a magnetic material. The magnetic material includes, for example, at least one selected from a Ni—Cu—Zn ferrite material, a Ni—Cu—Zn—Mg ferrite material, and a Ni—Cu ferrite material. The magnetic material constituting the element body **2** may contain an Fe alloy or the like. The element body **2** may be made of a nonmagnetic material. The nonmagnetic material includes, for example, at least one selected from a glass ceramic material and a dielectric material.

[0023] The element body **2** has, for example, a rectangular parallelepiped shape. The rectangular parallelepiped shape includes a rectangular parallelepiped shape in which corner portions and ridge portions are chamfered, and a rectangular parallelepiped shape in which the corner portions and the ridge portions are rounded. The rectangular parallelepiped shape includes a shape having a depression.

[0024] The element body **2** has a pair of side surfaces **3***a* opposite to each other, a pair of side surfaces **3***c* opposite to each other, and a pair of end surfaces **3***e* and **3***f* opposite to each other. Each of the side surfaces **3***a* and **3***c* is connected to the pair of end surfaces **3***e* and **3***f*. The pair of side surfaces **3***a*, the pair of side surfaces **3***c*, and the pair of end surfaces **3***e* and **3***f* have a rectangular shape. A direction in which the pair of side surfaces **3***a* is opposite to each other is a third direction D**3**. A direction in which the pair of side surfaces **3***c* is opposite to each other is a second direction D**2**. A direction in which the pair of end surfaces **3***e* and **3***f* is opposite to each other is a first direction D**1**. The first direction D**1** and the second direction D**2** intersect with each other. The third direction D**3** intersects the first direction D**1** and the second direction D**2**. In the example illustrated in the present embodiment, the first direction D**1**, the second direction D**2**, and the third direction D**3** are orthogonal to each other.

[0025] The coil component **1** is solder-mounted to an electronic device. The electronic device includes, for example, a circuit board or a multilayer coil component. In the coil component **1**, one side surface **3***a* faces the electronic device. The one side surface **3***a* is disposed so as to constitute a mounting surface. The one side surface **3***a* is the mounting surface. One side surface **3***c* of the pair of side surfaces **3***c* may be disposed to constitute the mounting surface. For example, when the side surface **3***a* constitutes a first side surface, the side surface **3***c* constitutes a second side surface.

[0026] For example, the third direction D**3** is orthogonal to each of the side surfaces **3***a*. The first direction D**1** is parallel to each of the side surfaces **3***a* and each of the side surfaces **3***c*. The second direction D**2** is orthogonal to each of the side surfaces **3***c*. In the present embodiment, the length of the element body **2** in the first direction D**1** is larger than the length of the element body **2** in the second direction D**2**, and is larger than the length of the element body **2** in the third direction D**3**. The first direction D**1** is a longitudinal direction of the element body **2**. The length of the element body **2** in the second direction D**2** and the length of the element body **2** in the third direction D**3** may be equal to each other. The length of the element body **2** in the first direction D**1** and the length of the element body **2** in the third direction D**3** may be different from each other. In the example illustrated in the present embodiment, the first direction D**1** corresponds to a Y-axis direction, the second direction D**2** corresponds to the X-axis direction, and the third direction D**3** corresponds to a Z-axis direction.

[0027] The length of the element body **2** in the third direction D**3** is the height of the element body **2**. The length of the element body **2** in the second direction D**2** is the width of the element body **2**. The length of the element body **2** in the first direction D**1** is the length of the element body **2**. In the present embodiment, a height T of the element body **2** is 0.05 to 1 mm, a width W of the element body **2** is 0.05 to 1 mm, and a length L of the element body **2** is 0.1 to 2 mm. For example, the height T of the element body **2** is 0.3 mm, the width W of the element body **2** is 0.2 mm, and the length L of the element body **2** is 0.4 mm.

[0028] The pair of side surfaces **3***c* extends in the third direction D**3** so as to couple the pair of side surfaces **3***a* to each other. The pair of side surfaces **3***c* also extends in the first direction D**1**. The pair of end surfaces **3***e* and **3***f* extends in the third direction D**3** so as to couple the pair of side surfaces **3***a* to each other. The pair of end surfaces **3***e* and **3***f* also extends in the second direction D**2**.

[0029] The element body **2** includes four ridge portions **3***g*, four ridge portions **3***i*, and four ridge portions **3***j*. The ridge portions **3***g* are located between the end surfaces **3***e* and **3***f* and the side surface **3***a*. The ridge portions **3***i* are located between the end surfaces **3***e* and **3***f* and the side surface **3***c*. The ridge portion **3***j* is located between the side surface **3***a* and the side surface **3***c*. In the present embodiment, each of the ridge portions **3***g*, **3***i*, and **3***j* is rounded so as to be curved. The element body **2** is subjected to so-called round chamfering. The end surfaces **3***e* and **3***f* and the side surface **3***a* are indirectly adjacent to each other via the ridge portions **3***g*. The end surfaces **3***e* and **3***f* and the side surface **3***c* are indirectly adjacent to each other via the ridge portions **3***i*. The side surface **3***a* and the side surface **3***c* are indirectly adjacent to each other via the ridge portion **3***j*.

[0030] The pair of external electrodes **5** and **6** is disposed on an outer surface of the element body **2** while being separated from each other. The pair of external electrodes **5** and **6** is opposite to each other in the Z-axis direction. The pair of external electrodes **5** and **6** is separated from each other in the Z-axis direction. The external electrodes **5** and **6** are each provided on the pair of side surfaces **3***c* and the pair of side surfaces **3***a*.

[0031] The pair of external electrodes **5** and **6** is formed by a known method. The pair of external electrodes **5** and **6** is made of, for example, a metal material. The metal material is, for example, copper, silver, gold, nickel, or chromium. The pair of external electrodes **5** and **6** is formed, for example, by subjecting an electrode layer to plating treatment. The electrode layer is made of, for example, a conductive paste. The conductive paste is applied by, for example, a dipping method, a printing method, or a transfer method. The electrode layer may be formed by, for example, a photolithography method. The plating treatment is, for example, electrolytic plating or electroless plating. By this plating treatment, a plating layer is formed on an outer surface of the conductive paste.

[0032] The external electrode **5** includes, for example, an end surface electrode portion **5***a* and a side surface electrode portion **5***b*. The end surface electrode portion **5***a* is provided on the end surface **3***e*. The side surface electrode portion **5***b* is provided on the pair of side surfaces **3***a* and the pair of side surfaces **3***c*. The side surface electrode portion **5***b* extends from the end surface **3***e* toward the end surface **3***f*. The external electrode **6** includes, for example, an end surface electrode portion **6***a* and a side surface electrode portion **6***b*. The end surface electrode portion **6***a* is provided on the end surface **3***f*. The side surface electrode portion **6***b* is provided on the pair of side surfaces **3***a* and the pair of side surfaces **3***c*. The side surface electrode portion **6***b* extends from the end surface **3***f* toward the end surface **3***e*. The side surface electrode portion **5***b* and the side surface electrode portion **6***b* cover, for example, a portion of each of the side surfaces **3***a* and a portion of each of the side surfaces **3***c*.

[0033] In the example illustrated in the present embodiment, the side surface electrode portion **5***b* of the external electrode **5** and the side surface electrode portion **6***b* of the external electrode **6** are formed in an annular shape. In each of the side surfaces **3***a* and **3***c*, a region covered with the side surface electrode portion **5***b* and the side surface electrode portion **6***b* has, for example, a rectangular shape when viewed from the X-axis direction or the Z-axis direction. In the present specification, the term “couple” means connecting in a directly contacted state. The phrase “directly contact” means that members are connected to each other without interposing another member illustrated in the present specification. The phrase “directly contact” does not exclude being connected via a member not explicitly described herein. The term “connect” includes not only a directly contacted state but also a physically separated and electrically connected state unless otherwise specified.

[0034] As illustrated in FIGS. **2** to **4**, the coil component **1** further includes a coil CL. The coil CL is disposed inside the element body **2**. FIG. **2** is a partial cross-sectional view of the coil component in the YZ-axis plane. FIG. **3** is a partial cross-sectional view of the coil component in the XY-axis plane. FIG. **4** is a partial cross-sectional view of the coil component in the XZ-axis plane.

[0035] The coil CL electrically connects the external electrode **5** and the external electrode **6**. The coil CL forms a coil axis CA along an opposite direction of the pair of side surfaces **3***c*. The coil axis CA extends, for example, in the X-axis direction. That is, the X-axis direction corresponds to a coil axis direction. The coil CL includes ends CL**1** and CL**2** separated from each other. For example, the coil CL has a spiral structure in which the coil advances clockwise from the end CL**1** toward the end CL**2** when viewed from the X-axis direction.

[0036] The end CL**1** of the coil CL is connected to the external electrode **5**. The end CL**2** of the coil CL is connected to the external electrode **6**. A portion of the coil CL overlaps the side surface electrode portions **5***b* and **6***b* when viewed along the Z-axis direction. The coil CL includes a plurality of coil conductors **10**. The plurality of coil conductors **10** correspond to internal conductors. The plurality of coil conductors **10** include a plurality of coil conductors **11**, **12**, **13**, **14**, **15**, **21**, **22**, **23**, and **24**. In the example illustrated in the present embodiment, the coil CL is a triple-wound coil and includes three each of the coil conductors **11**, **12**, **13**, **14**, **15**, **21**, **22**, **23**, and **24**. Each of the plurality of coil conductors **10** is coupled at an end thereof to another coil conductor **10** of the plurality of coil conductors **10**. In the example illustrated in the present embodiment, a region R surrounded by the plurality of coil conductors **10** has an octagonal shape when viewed along a direction of the coil axis CA, that is, along the X-axis direction.

[0037] The plurality of coil conductors **11**, **12**, **13**, **14**, **15**, **21**, **22**, **23**, and **24** are electrically connected to each other. In the example illustrated in the present embodiment, the plurality of coil conductors **11**, **12**, **13**, **14**, and **15** are provided in the same layer. The plurality of coil conductors **11**, **12**, **13**, **14**, and **15** form a coil conductor layer **17**. The plurality of coil conductors **21**, **22**, **23**, and **24** are provided in the same layer. The plurality of coil conductors **21**, **22**, **23**, and **24** form a coil conductor layer **27**. A plurality of the coil conductor layers **17** and a plurality of the coil conductor layers **27** are provided at different positions in the X-axis direction.

[0038] The coil conductor **11** includes a pair of ends **11***a* and **11***b*. The coil conductor **12** includes a pair of ends **12***a* and **12***b*. The coil conductor **13** includes a pair of ends **13***a* and **13***b*. The coil conductor **14** includes a pair of ends **14***a* and **14***b*. The coil conductor **15** includes a pair of ends **15***a* and **15***b*.

[0039] The end **11***a* of the coil conductor **11** is exposed from the end surface **3***e* of the element body **2**, and is coupled to the end surface electrode portion **5***a* of the external electrode **5** at the end surface **3***e*. The end **11***b* of the coil conductor **11** and the end **12***a* of the coil conductor **12** are coupled to each other. The end **12***b* of the coil conductor **12** and the end **13***a* of the coil conductor **13** are coupled to each other. The end **13***b* of the coil conductor **13** and the end **14***a* of the coil conductor **14** are coupled to each other. The end **14***b* of the coil conductor **14** and the end **15***a* of the coil conductor **15** are coupled to each other.

[0040] The coil conductor **11** extends along the Y-axis direction. When viewed along the Z-axis direction orthogonal to the direction of the coil axis CA of the coil CL and the Y-axis direction, the coil conductor **11** does not overlap a portion **6***c* which is closest to the coil conductor **11** in the side surface electrode portion **6***b*. The portion **6***c* corresponds to an edge of the side surface electrode portion **6***b* opposite to the side surface electrode portion **5***b* in the Y axis direction, and the coil conductor **11** does not overlap a boundary BD formed by the edge. The coil conductor **11** is separated from the portion **6***c* which is closest to the coil conductor **11** in the side surface electrode portion **6***b* in the Y-axis direction. In the example illustrated in the present embodiment, when viewed along the Z-axis direction, the coil conductor **11** does not overlap the side surface electrode portion **6***b* provided on the side surface **3***a* close to the coil conductor **11** in the pair of side surfaces

3*a*. In the example illustrated in the present embodiment, when viewed along the Z-axis direction, the coil conductor **11** does not overlap the side surface electrode portion **6***b* provided on each of the pair of side surfaces **3***a*. The coil conductor **11** is separated from the side surface electrode portion **6***b* provided on each of the pair of side surfaces **3***a* in the Y-axis direction.

[0041] The coil conductor **12** extends from the coil conductor **11** toward the end surface **3***f* such that the shortest distance from the side surface **3***a* to the coil conductor **12** increases as the coil conductor **12** approaches the end surface **3***f*. In other words, the coil conductor **12** extends from the coil conductor **13** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **12** decreases as the distance from the end surface **3***f* increases. A portion of the coil conductor **12** overlaps the side surface electrode portion **6***b* when viewed along the Z-axis direction. The boundary BD passes through the coil conductor **12**. The coil conductor **12** is orthogonal to the X-axis direction and extends in a direction inclined with respect to the Y-axis direction and the Z-axis direction.

[0042] The coil conductor **13** is connected to the end **12***b* different from the end **12***a* connected to the coil conductor **11**, in the pair of ends **12***a* and **12***b* of the coil conductor **12**. The coil conductor **13** extends in the Z-axis direction. A longitudinal direction of the coil conductor **13** is orthogonal to the pair of side surfaces **3***a*. The entire coil conductor **13** overlaps the side surface electrode portion **6***b* when viewed along the Z-axis direction.

[0043] The coil conductor **14** extends from the coil conductor **15** toward the end surface **3***f* such that the shortest distance from the side surface **3***a* to the coil conductor **14** increases as the coil conductor **14** approaches the end surface **3***f*. In other words, the coil conductor **14** extends from the coil conductor **13** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **14** decreases as the distance from the end surface **3***f* increases. A portion of the coil conductor **14** overlaps the side surface electrode portion **6***b* when viewed along the Z-axis direction. The coil conductor **14** is orthogonal to the X-axis direction and extends in a direction inclined with respect to the Y-axis direction and the Z-axis direction.

[0044] The coil conductor **15** extends along the Y-axis direction. When viewed along the Z-axis direction orthogonal to the direction of the coil axis CA of the coil CL and the Y-axis direction, the coil conductor **15** does not overlap a portion **5***c* which is closest to the coil conductor **15** in the side surface electrode portion **5***b*. When viewed along the Z-axis direction, the coil conductor **15** does not overlap the portion **6***c* which is closest to the coil conductor **15** in the side surface electrode portion **6***b*. The coil conductor **15** is separated from the portion **5***c* which is closest to the coil conductor **15** in the side surface electrode portion **5***b* in the Y-axis direction. The coil conductor **15** is separated from the portion **6***c* which is closest to the coil conductor **15** in the side surface electrode portion **6***c* in the Y-axis direction. The coil conductor **15** is connected to the end **14***b* different from the end **14***a* connected to the coil conductor **13**, in the pair of ends **14***a* and **14***b* of the coil conductor **14**.

[0045] In the example illustrated in the present embodiment, when viewed along the Z-axis direction, the coil conductor **15** does not overlap the side surface electrode portion **5***b* provided on the side surface **3***a* close to the coil conductor **15** in the pair of side surfaces **3***a*. When viewed along the Z-axis direction, the coil conductor **15** does not overlap the side surface electrode portion **6***b* provided on the side surface **3***a* close to the coil conductor **15** in the pair of side surfaces **3***a*. In the example illustrated in the present embodiment, when viewed along the Z-axis direction, the coil conductor **15** does not overlap the side surface electrode portion **5***b* provided on each of the pair of side surfaces **3***a*. When viewed along the Z-axis direction, the coil conductor **15** does not overlap the side surface electrode portion **6***b* provided on each of the pair of side surfaces **3***a*. The coil conductor **15** is separated from the side surface electrode portions **5***b* and **6***b* provided on each of the pair of side surfaces **3***a* in the Y-axis direction.

[0046] The coil conductor **21** includes a pair of ends **21***a* and **21***b*. The plurality of coil conductors **22** include a pair of ends **22***a* and **22***b*. The plurality of coil conductors **23** include a pair of ends

23*a* and **23***b*. The plurality of coil conductors **24** include a pair of ends **24***a* and **24***b*.

[0047] The end **21***a* of the coil conductor **21** is exposed from the end surface **3***f* opposite to the end surface **3***e* from which the end **11***a* is exposed, in the pair of end surfaces **3***e* and **3***f*, and is coupled to the end surface electrode portion **6***a* of the external electrode **6** at the end surface **3***f*. The end **21***b* of the coil conductor **21** and the end **22***a* of the coil conductor **22** are coupled to each other. The end **22***b* of the coil conductor **22** and the end **23***a* of the coil conductor **23** are coupled to each other. The end **23***b* of the coil conductor **23** and the end **24***a* of the coil conductor **24** are coupled to each other. The end **24***b* of the coil conductor **24** and the end **15***b* of the coil conductor **15** are connected to each other via a via **31**.

[0048] The coil conductor **21** extends along the Y-axis direction. When viewed along the Z-axis direction orthogonal to the direction of the coil axis CA of the coil CL and the Y-axis direction, the coil conductor **21** does not overlap the portion **5***c* which is closest to the coil conductor **21** in the side surface electrode portion **5***b*. The coil conductor **21** is separated from the portion **5***c* which is closest to the coil conductor **21** in the side surface electrode portion **5***b* in the Y-axis direction. In the example illustrated in the present embodiment, when viewed along the Z-axis direction, the coil conductor **21** does not overlap the side surface electrode portion **5***b* provided on the side surface **3***a* close to the coil conductor **21** in the pair of side surfaces **3***a*. In the example illustrated in the present embodiment, when viewed along the Z-axis direction, the coil conductor **21** does not overlap the side surface electrode portion **5***b* provided on each of the pair of side surfaces **3***a*. The coil conductor **21** is separated from the side surface electrode portion **5***b* provided on each of the pair of side surfaces **3***a* in the Y-axis direction.

[0049] The coil conductor **22** extends from the coil conductor **21** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **22** increases as the coil conductor **22** approaches the end surface **3***e*. In other words, the coil conductor **22** extends from the coil conductor **23** toward the end surface **3***f* such that the shortest distance from the side surface **3***a* to the coil conductor **22** decreases as the distance from the end surface **3***e* increases. A portion of the coil conductor **22** overlaps the side surface electrode portion **5***b* when viewed along the Z-axis direction. The coil conductor **22** is orthogonal to the X-axis direction and extends in a direction inclined with respect to the Y-axis direction and the Z-axis direction.

[0050] The coil conductor **23** is connected to the end **22***b* of the pair of ends **22***a* and **22***b* of the coil conductor **22**. The coil conductor **23** extends in the Z-axis direction. A longitudinal direction of the coil conductor **23** is orthogonal to the pair of side surfaces **3***a*. The entire coil conductor **23** overlaps the side surface electrode portion **5***b* when viewed along the Z-axis direction.

[0051] The coil conductor **24** extends from the coil conductor **15** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **24** increases as the coil conductor **24** approaches the end surface **3***e*. In other words, the coil conductor **24** extends from the coil conductor **23** toward the end surface **3***f* such that the shortest distance from the side surface **3***a* to the coil conductor **24** decreases as the distance from the end surface **3***e* increases. A portion of the coil conductor **24** overlaps the side surface electrode portion **5***b* when viewed along the Z-axis direction. The coil conductor **24** is orthogonal to the X-axis direction and extends in a direction inclined with respect to the Y-axis direction and the Z-axis direction.

[0052] The coil CL is made of a conductive material. An internal conductor layer provided inside the coil component **1** is made of a conductive material. The conductive material contains, for example, at least one selected from Ag and Pd.

[0053] Next, a coil component **1**A in a modification of the present embodiment will be described with reference to FIGS. **5** to **8**D. FIG. **5** is a partial cross-sectional view of the coil component in the YZ-axis plane in the modification of the present embodiment. FIG. **6** is a partial cross-sectional view of the coil component in the XY-axis plane. FIG. **7** is a partial cross-sectional view of the coil component in the XZ-axis plane. FIGS. **8**A to **8**D illustrate a coil conductor in each layer. The present modification is generally similar to or the same as the coil component **1** in the above-described embodiment. The present modification is different from the above-described embodiment in that the configuration of the coil CL is different. Hereinafter, differences between the above-described embodiment and the modification will be mainly described.

[0054] The coil CL includes a plurality of coil conductors **40**. The plurality of coil conductors **40** correspond to internal conductors. The plurality of coil conductors **40** include a plurality of coil conductors **41**, **42**, **43**, **44**, **51**, **52**, **53**, **54**, and **55** and a plurality of coil conductors **61**, **62**, **63**, **64**, **65**, **71**, **72**, **73**, **74**, **75**, and **76**. In the present modification, the coil CL is a single-wound coil. The plurality of coil conductors **41**, **42**, **43**, **44**, **51**, **52**, **53**, **54**, and **55** and the plurality of coil conductors **61**, **62**, **63**, **64**, **65** **71**, **72**, **73**, **74**, **75**, and **76** are electrically connected to each other.

[0055] In the present modification, the plurality of coil conductors **41**, **42**, **43**, and **44** are provided in the same layer. The plurality of coil conductors **41**, **42**,**43**, and **44** form a coil conductor layer **47**. The plurality of coil conductors **51**, **52**, **53**, **54**, and **55** are provided in the same layer. The plurality of coil conductors **51**, **52**, **53**, **54**, and **55** form a coil conductor layer **57**. The plurality of coil conductors **61**, **62**, **63**, **64**, and **65** are provided in the same layer. The plurality of coil conductors **61**, **62**, **63**, **64**, and **65** form a coil conductor layer **67**. The plurality of coil conductors **71**, **72**, **73**, **74**, **75**, and **76** are provided in the same layer. The plurality of coil conductors **71**, **72**, **73**, **74**, **75**, and **76** form a coil conductor layer **77**. The coil conductor layer **47**, the coil conductor layer **57**, the coil conductor layer **67**, and the coil conductor layer **77** are provided at different positions in the X-axis direction.

[0056] The coil conductor **41** includes a pair of ends **41***a* and **41***b*. The coil conductor **42** includes a pair of ends **42***a* and **42***b*. The coil conductor **43** includes a pair of ends **43***a* and **43***b*. The coil conductor **44** includes a pair of ends **44***a* and **44***b*.

[0057] The end **41***a* of the coil conductor **41** is exposed from the end surface **3***e* of the element body **2**, and is coupled to the end surface electrode portion **5***a* of the external electrode **5** at the end surface **3***e*. The end **41***b* of the coil conductor **41** and the end **42***a* of the coil conductor **42** are coupled to each other. The end **42***b* of the coil conductor **42** and the end **43***a* of the coil conductor **43** are coupled to each other. The end **43***b* of the coil conductor **43** and the end **44***a* of the coil conductor **44** are coupled to each other.

[0058] The coil conductor **41** extends along the Y-axis direction. When viewed along the Z-axis direction orthogonal to the direction of the coil axis CA of the coil CL and the Y-axis direction, the coil conductor **41** does not overlap the portion **6***c* which is closest to the coil conductor **41** in the side surface electrode portion **6***b*. The coil conductor **41** is separated from the portion **6***c* which is closest to the coil conductor **41** in the side surface electrode portion **6***b* in the Y-axis direction. In the example illustrated in the present embodiment, when viewed along the Z-axis direction, the coil conductor **41** does not overlap the side surface electrode portion **6***b* provided on the side surface **3***a* close to the coil conductor **41** in the pair of side surfaces **3***a*. In the example illustrated in the present embodiment, when viewed along the Z-axis direction, the coil conductor **41** does not overlap the side surface electrode portion **6***b* provided on each of the pair of side surfaces **3***a*. The coil conductor **41** is separated from the side surface electrode portion **6***b* provided on each of the pair of side surfaces **3***a* in the Y-axis direction.

[0059] The coil conductor **42** extends from the coil conductor **41** toward the end surface **3***f* such that the shortest distance from the side surface **3***a* to the coil conductor **42** increases as the coil conductor **42** approaches the end surface **3***f*. In other words, the coil conductor **42** extends from the coil conductor **43** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **42** decreases as the distance from the end surface **3***f* increases. A portion of the coil conductor **42** overlaps the side surface electrode portion **6***b* when viewed along the Z-axis direction. The coil conductor **42** is orthogonal to the X-axis direction and extends in a direction inclined with respect to the Y-axis direction and the Z-axis direction.

[0060] The coil conductor **43** is connected to the end **42***b* different from the end **42***a* connected to the coil conductor **44**, in the pair of ends **42***a* and **42***b* of the coil conductor **42**. The coil conductor

43 extends in the Z-axis direction. A longitudinal direction of the coil conductor **43** is orthogonal to the pair of side surfaces **3***a*. The entire coil conductor **43** overlaps the side surface electrode portion **6***b* when viewed along the Z-axis direction.

[0061] The coil conductor **44** extends from the coil conductor **51** toward the end surface **3***f* such that the shortest distance from the side surface **3***a* to the coil conductor **44** increases as the coil conductor **44** approaches the end surface **3***f*. In other words, the coil conductor **44** extends from the coil conductor **43** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **44** decreases as the distance from the end surface **3***f* increases. A portion of the coil conductor **44** overlaps the side surface electrode portion **6***b* when viewed along the Z-axis direction. The coil conductor **44** is orthogonal to the X-axis direction and extends in a direction inclined with respect to the Y-axis direction and the Z-axis direction.

[0062] The coil conductor **51** includes a pair of ends **51***a* and **51***b*. The coil conductor **52** includes a pair of ends **52***a* and **52***b*. The coil conductor **53** includes a pair of ends **53***a* and **53***b*. The coil conductor **54** includes a pair of ends **54***a* and **54***b*. The coil conductor **55** includes a pair of ends **55***a* and **55***b*.

[0063] The end **51***b* of the coil conductor **51** and the end **52***a* of the coil conductor **52** are coupled to each other. The end **52***b* of the coil conductor **52** and the end **53***a* of the coil conductor **53** are coupled to each other. The end **53***b* of the coil conductor **53** and the end **54***a* of the coil conductor **54** are coupled to each other. The end **54***b* of the coil conductor **54** and the end **55***a* of the coil conductor **55** are coupled to each other.

[0064] The coil conductor **51** extends in the same direction as the coil conductor **44**. When viewed along the X-axis direction, the coil conductor **51** and the coil conductor **44** are located on the same straight line. The end **44***b* of the coil conductor **44** and the end **51***a* of the coil conductor **51** are connected to each other via a via **81**.

[0065] The coil conductor **52** extends along the Y-axis direction. When viewed along the Z-axis direction orthogonal to the direction of the coil axis CA of the coil CL and the Y-axis direction, the coil conductor **52** does not overlap the portion **5***c* which is closest to the coil conductor **52** in the side surface electrode portion **5***b*. When viewed along the Z-axis direction, the coil conductor **52** does not overlap the portion **6***c* which is closest to the coil conductor **52** in the side surface electrode portion **6***b*. The coil conductor **52** is separated from the portion **5***c* which is closest to the coil conductor **52** in the side surface electrode portion **5***b* in the Y-axis direction. The coil conductor **52** is separated from the portion **6***c* which is closest to the coil conductor **52** in the side surface electrode portion **6***b* in the Y-axis direction. The coil conductor **52** is connected to the end **51***b* of the coil conductor **51**.

[0066] In the present modification, when viewed along the Z-axis direction, the coil conductor **52** does not overlap the side surface electrode portion **5***b* provided on the side surface **3***a* close to the coil conductor **52** in the pair of side surfaces **3***a*. When viewed along the Z-axis direction, the coil conductor **52** does not overlap the side surface electrode portion **6***b* provided on the side surface **3***a* close to the coil conductor **52** in the pair of side surfaces **3***a*. In the present modification, when viewed along the Z-axis direction, the coil conductor **52** does not overlap the side surface electrode portion **5***b* provided on each of the pair of side surfaces **3***a*. When viewed along the Z-axis direction, the coil conductor **52** does not overlap the side surface electrode portion **6***b* provided on each of the pair of side surfaces **3***a*. The coil conductor **52** is separated from the side surface electrode portions **5***b* and **6***b* provided on each of the pair of side surfaces **3***a* in the Y-axis direction.

[0067] The coil conductor **53** extends from the coil conductor **52** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **53** increases as the coil conductor **53** approaches the end surface **3***e*. In other words, the coil conductor **53** extends from the coil conductor **54** toward the end surface **3***f* such that the shortest distance from the side surface **3***a* to the coil conductor **53** decreases as the distance from the end surface **3***e* increases. A portion of the coil conductor **53** overlaps the side surface electrode portion **5***b* when viewed along the Z-axis direction. The coil conductor **53** is orthogonal to the X-axis direction and extends in a direction inclined with respect to the Y-axis direction and the Z-axis direction.

[0068] The coil conductor **54** is connected to the end **53***b* of the pair of ends **53***a* and **53***b* of the coil conductor **53**. The coil conductor **54** extends in the Z-axis direction. A longitudinal direction of the coil conductor **54** is orthogonal to the pair of side surfaces **3***a*. The entire coil conductor **54** overlaps the side surface electrode portion **5***b* when viewed along the Z-axis direction.

[0069] The coil conductor **55** extends from the coil conductor **61** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **55** increases as the coil conductor **55** approaches the end surface **3***e*. In other words, the coil conductor **55** extends from the coil conductor **54** toward the end surface **3***f* such that the shortest distance from the side surface **3***a* to the coil conductor **55** decreases as the distance from the end surface **3***e* increases. A portion of the coil conductor **55** overlaps the side surface electrode portion **5***b* when viewed along the Z-axis direction. The coil conductor **55** is orthogonal to the X-axis direction and extends in a direction inclined with respect to the Y-axis direction and the Z-axis direction.

[0070] The coil conductor **61** includes a pair of ends **61***a* and **61***b*. The coil conductor **62** includes a pair of ends **62***a* and **62***b*. The coil conductor **63** includes a pair of ends **63***a* and **63***b*. The coil conductor **64** includes a pair of ends **64***a* and **64***b*. The coil conductor **65** includes a pair of ends **65***a* and **65***b*.

[0071] The end **61***b* of the coil conductor **61** and the end **62***a* of the coil conductor **62** are coupled to each other. The end **62***b* of the coil conductor **62** and the end **63***a* of the coil conductor **63** are coupled to each other. The end **63***b* of the coil conductor **63** and the end **64***a* of the coil conductor **64** are coupled to each other. The end **64***b* of the coil conductor **64** and the end **65***a* of the coil conductor **65** are coupled to each other.

[0072] The coil conductor **61** extends in the same direction as the coil conductor **55**. When viewed along the X-axis direction, the coil conductor **61** and the coil conductor **55** are located on the same straight line. The end **55***b* of the coil conductor **55** and the end **61***a* of the coil conductor **61** are connected to each other via a via **91**.

[0073] The coil conductor **62** extends along the Y-axis direction. When viewed along the Z-axis direction orthogonal to the direction of the coil axis CA of the coil CL and the Y-axis direction, the coil conductor **62** does not overlap the portion **5***c* which is closest to the coil conductor **62** in the side surface electrode portion **5***b*. When viewed along the Z-axis direction, the coil conductor **62** does not overlap the portion **6***c* which is closest to the coil conductor **62** in the side surface electrode portion **6***b*. The coil conductor **62** is separated from the portion **5***c* which is closest to the coil conductor **52** in the side surface electrode portion **5***b* in the Y-axis direction. The coil conductor **62** is separated from the portion **6***c* which is closest to the coil conductor **62** in the side surface electrode portion **6***b* in the Y-axis direction. The coil conductor **62** is connected to the end **61***b* of the coil conductor **61**.

[0074] In the present modification, when viewed along the Z-axis direction, the coil conductor **62** does not overlap the side surface electrode portion **5***b* provided on the side surface **3***a* close to the coil conductor **62** in the pair of side surfaces **3***a*. When viewed along the Z-axis direction, the coil conductor **62** does not overlap the side surface electrode portion **6***b* provided on the side surface **3***a* close to the coil conductor **62** in the pair of side surfaces **3***a*. In the present modification, when viewed along the Z-axis direction, the coil conductor **62** does not overlap the side surface electrode portion **5***b* provided on each of the pair of side surfaces **3***a*. When viewed along the Z-axis direction, the coil conductor **62** does not overlap the side surface electrode portion **6***b* provided on each of the pair of side surfaces **3***a*. The coil conductor **62** is separated from the side surface electrode portions **5***b* and **6***b* provided on each of the pair of side surfaces **3***a* in the Y-axis direction.

[0075] The coil conductor **63** extends from the coil conductor **62** toward the end surface **3***f* such that the shortest distance from the side surface **3***a* to the coil conductor **63** increases as the coil conductor **63** approaches the end surface **3***f*. In other words, the coil conductor **63** extends from the coil conductor **64** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **63** decreases as the distance from the end surface **3***f* increases. A portion of the coil conductor **63** overlaps the side surface electrode portion **6***b* when viewed along the Z-axis direction. The coil conductor **63** is orthogonal to the X-axis direction and extends in a direction inclined with respect to the Y-axis direction and the Z-axis direction.

[0076] The coil conductor **64** is connected to the end **63***b* of the coil conductor **63**. The coil conductor **64** extends in the Z-axis direction. A longitudinal direction of the coil conductor **64** is orthogonal to the pair of side surfaces **3***a*. The entire coil conductor **64** overlaps the side surface electrode portion **6***b* when viewed along the Z-axis direction.

[0077] The coil conductor **65** extends from the coil conductor **21** toward the end surface **3***f* such that the shortest distance from the side surface **3***a* to the coil conductor **65** increases as the coil conductor **65** approaches the end surface **3***f*. In other words, the coil conductor **65** extends from the coil conductor **64** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **65** decreases as the distance from the end surface **3***f* increases. A portion of the coil conductor **65** overlaps the side surface electrode portion **6***b* when viewed along the Z-axis direction. The coil conductor **65** is orthogonal to the X-axis direction and extends in a direction inclined with respect to the Y-axis direction and the Z-axis direction.

[0078] The coil conductor **71** includes a pair of ends **71***a* and **71***b*. The coil conductor **72** includes a pair of ends **72***a* and **72***b*. The coil conductor **73** includes a pair of ends **73***a* and **73***b*. The coil conductor **74** includes a pair of ends **74***a* and **74***b*. The coil conductor **75** includes a pair of ends **75***a* and **75***b*. The coil conductor **76** includes a pair of ends **76***a* and **76***b*.

[0079] The end **71***b* of the coil conductor **71** and the end **72***a* of the coil conductor **72** are coupled to each other. The end **72***b* of the coil conductor **72** and the end **73***a* of the coil conductor **73** are coupled to each other. The end **73***b* of the coil conductor **73** and the end **74***a* of the coil conductor **74** are coupled to each other. The end **74***b* of the coil conductor **74** and the end **75***a* of the coil conductor **75** are coupled to each other. The end **75***b* of the coil conductor **75** and the end **76***a* of the coil conductor **76** are coupled to each other.

[0080] The coil conductor **71** extends in the same direction as the coil conductor **65**. When viewed along the X-axis direction, the coil conductor **71** and the coil conductor **65** are located on the same straight line. The end **65***b* of the coil conductor **65** and the end **71***a* of the coil conductor **71** are connected to each other via the via **81**.

[0081] The coil conductor **72** has the same shape as the coil conductor **52** and has a configuration similar to that of the coil conductor **52**. The coil conductor **72** is connected to the end **71***b* of the coil conductor **71**. The coil conductor **73** has the same shape as the coil conductor **53** and has a configuration similar to that of the coil conductor **53**. The coil conductor **73** is connected to the end **72***b* of the coil conductor **72**. The coil conductor **74** has the same shape as the coil conductor **54** and has a configuration similar to that of the coil conductor **54**. The coil conductor **74** is connected to the end **73***b* of the coil conductor **73**.

[0082] The coil conductor **75** extends from the coil conductor **76** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **75** increases as the coil conductor **75** approaches the end surface **3***e*. In other words, the coil conductor **75** extends from the coil conductor **74** toward the end surface **3***f* such that the shortest distance from the side surface **3***a* to the coil conductor **75** decreases as the distance from the end surface **3***e* increases. A portion of the coil conductor **75** overlaps the side surface electrode portion **5***b* when viewed along the Z-axis direction. The coil conductor **75** is orthogonal to the X-axis direction and extends in a direction inclined with respect to the Y-axis direction and the Z-axis direction.

[0083] The coil conductor **76** extends along the Y-axis direction. When viewed along the Z-axis direction, the coil conductor **76** does not overlap the portion **5***c* which is closest to the coil conductor **76** in the side surface electrode portion **5***b*. The coil conductor **76** is separated from the portion **5***c* which is closest to the coil conductor **76** in the side surface electrode portion **5***b* in the Y-

axis direction. In the present modification, when viewed along the Z-axis direction, the coil conductor **76** does not overlap the side surface electrode portion **5***b* provided on the side surface **3***a* close to the coil conductor **76** in the pair of side surfaces **3***a*.
[0084] Next, effects of the coil components **1** and **1**A in the present embodiment and the modification will be described.
[0085] FIG. **9** is a graph illustrating frequency characteristics of a Q value. Data DA**1** illustrates a frequency characteristic of the Q value in a normal coil component. Data DA**2** illustrates a frequency characteristic of the Q value in a case where a coil is short-circuited to an external electrode in a portion other than an end. Data DA**3** illustrates data in a configuration in which a desired Q value is secured when an electrical signal in a relatively low frequency band conducts through the coil, and the Q value decreases when an electrical signal in a relatively high frequency band conducts through the coil. When there is a position where a potential difference between the coil and the external electrode is relatively large and the distance is relatively short, the Q value decreases for the electric signal having a relatively low frequency as illustrated in the data DA**3**.
[0086] In the coil component **1**, since the coil conductor **11** connected to the external electrode **5** is separated from the external electrode **6** when viewed along the Y-axis direction, the Q value can be secured even for a signal in a relatively high frequency band while compactness is achieved. Since the coil conductor **12** is connected to the coil conductor **11** and extends from the coil conductor **11** toward the end surface **3***e* such that the shortest distance from the side surface **3***a* to the coil conductor **12** increases as the coil conductor **12** approaches the end surface **3***e*, the area of the region R surrounded by the coil CL can be secured when viewed along the direction of the coil axis CA. As a result, it is possible to secure inductance while achieving compactness. The coil component **1**A also has a similar configuration.
[0087] In the example illustrated in the present embodiment, the coil CL further includes the coil conductor **13** connected to the end **12***b* of the coil conductor **12**. The coil conductor **13** extends along the Z-axis direction. In this case, when viewed along the direction of the coil axis CA, a larger area of the region R surrounded by the coil CL can be secured. As a result, it is possible to more reliably secure inductance while achieving compactness. The coil component **1**A also has a similar configuration.
[0088] In the example illustrated in the present embodiment, a portion of the coil CL overlaps the side surface electrode portion **6***b* when viewed along the Z-axis direction. In this case, when viewed along the direction of the coil axis CA, a larger area of the region R surrounded by the coil CL can be secured. As a result, it is possible to more reliably secure inductance while achieving compactness. The coil component **1**A also has a similar configuration.
[0089] In the example illustrated in the present embodiment, the region R surrounded by the plurality of coil conductors **11** has an octagonal shape when viewed along the direction of the coil axis CA. In this case, when viewed along the direction of the coil axis CA, a larger area of the region R surrounded by the coil CL can be secured. As a result, it is possible to more reliably secure inductance while achieving compactness.
[0090] Although the embodiment and modification of the present invention have been described above, the present invention is not necessarily limited to the above-described embodiment, and various modifications can be made without departing from the gist thereof.
[0091] For example, in the examples illustrated in the embodiment and the modification described above, a case where the region R has an octagonal shape when viewed along the X-axis direction has been described. However, the shape of the region R is not limited thereto. For example, when viewed along the X-axis direction, the region R may have a hexagonal shape or a polygonal shape having 10 or more sides. The region R may not have a strictly polygonal shape, and the coil conductor **10** constituting the side may not have a linear shape.
[0092] As understood from the description of the above-described embodiment, the present specification includes disclosure of the following aspects.

(Appendix 1)
[0093] A coil component including: [0094] an element body including first and second end surfaces opposite to each other in a first direction and a side surface connected to the first and second end surfaces; [0095] a first external electrode including a first end surface electrode portion provided on the first end surface and a first side surface electrode portion provided on the side surface and extending from the first end surface toward the second end surface; [0096] a second external electrode including a second end surface electrode portion provided on the second end surface and a second side surface electrode portion provided on the side surface and extending from the second end surface toward the first end surface; and [0097] a coil disposed inside the element body, coupled to the first end surface electrode portion at the first end surface, and coupled to the second end surface electrode portion at the second end surface, in which [0098] the coil includes a plurality of coil conductors each including a pair of ends, [0099] each of the plurality of coil conductors is connected at the end to another coil conductor of the plurality of coil conductors, [0100] the plurality of coil conductors include a first coil conductor extending along the first direction and a second coil conductor connected to the first coil conductor and extending from the first coil conductor toward the second end surface such that a shortest distance from the side surface to the second coil conductor increases as the second coil conductor approaches the second end surface, and [0101] when viewed along a second direction orthogonal to a coil axis direction of the coil and the first direction, the first coil conductor does not overlap a portion which is closest to the first coil conductor in the second side surface electrode portion.
(Appendix 2)
[0102] The coil component according to Appendix 1, in which the coil further includes a third coil conductor connected to an end different from an end connected to the first coil conductor, in the pair of ends of the second coil conductor, and [0103] the third coil conductor extends in the second direction.
(Appendix 3)
[0104] The coil component according to Appendix 1 or 2, in which a portion of the coil overlaps the second side surface electrode portion when viewed along the second direction.
(Appendix 4)
[0105] The coil component according to any one of Appendixes 3 to 1, in which a region surrounded by the plurality of coil conductors has an octagonal shape when viewed along the coil axis direction.
Reference Signs List
[0106] **1** COIL COMPONENT [0107] **2** ELEMENT BODY [0108] **3***a*, **3***c* SIDE SURFACE [0109] **3***e*, **3***f* END SURFACE [0110] **5** **5**,**6** EXTERNAL ELECTRODE [0111] **5***a*, **6***a* END SURFACE ELECTRODE PORTION [0112] **5***b*, **6***b* SIDE SURFACE ELECTRODE PORTION [0113] **10**, **11**, **12**, **13** COIL CONDUCTOR [0114] **6***c* PORTION [0115] **10** CL COIL [0116] CA COIL AXIS [0117] R REGION

Claims

1. A coil component comprising: an element body including first and second end surfaces opposite to each other in a first direction and a side surface connected to the first and second end surfaces; a first external electrode including a first end surface electrode portion provided on the first end surface and a first side surface electrode portion provided on the side surface and extending from the first end surface toward the second end surface; a second external electrode including a second end surface electrode portion provided on the second end surface and a second side surface electrode portion provided on the side surface and extending from the second end surface toward the first end surface; and a coil disposed inside the element body, coupled to the first end surface electrode portion at the first end surface, and coupled to the second end surface electrode portion at the second end surface, wherein the coil includes a plurality of coil conductors each including a pair of ends, each of the plurality of coil conductors is connected at the end to another coil conductor of the plurality of coil conductors, the plurality of coil conductors include a first coil conductor extending along the first direction and a second coil conductor connected to the first coil conductor and extending from the first coil conductor toward the second end surface such that a shortest distance from the side surface to the second coil conductor increases as the second coil conductor approaches the second end surface, and when viewed along a second direction orthogonal to a direction of a coil axis of the coil and the first direction, the first coil conductor does not overlap a portion which is closest to the first coil conductor in the second side surface electrode portion.

2. The coil component according to claim 1, wherein the coil further includes a third coil conductor connected to an end different from an end connected to the first coil conductor, in the pair of ends of the second coil conductor, and the third coil conductor extends in the second direction.

3. The coil component according to claim 1, wherein a portion of the coil overlaps the second side surface electrode portion when viewed along the second direction.

4. The coil component according to claim 1, wherein a region surrounded by the plurality of coil conductors has an octagonal shape when viewed along the direction of the coil axis.